

Project Design Phase

Proposed Solution Template

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:

The proposed solution summarizes how the Metro Ticket Generating System addresses the identified commuter and metro-operations problems through a ServiceNow-based digital ticketing workflow.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Metro passengers may depend on manual or paper-based ticketing processes that can involve queues, repeated data entry, slower validation, and limited visibility of booking status. Metro operations teams also need accurate and centralized ticket records, reliable fare processing, and a simple way to validate digital tickets. The problem is therefore to create a faster, more reliable, and maintainable digital metro ticketing workflow that reduces manual intervention while improving passenger convenience and operational accuracy.
2	Idea / Solution Description	Develop a Metro Ticket Generating System on ServiceNow. Passengers use a Service Catalog item to select Source, Destination, Passenger Type, and Number of Tickets. The system applies fare logic, processes the request through Flow Designer, maps booking data into a custom Metro Database table, and generates a QR-based digital ticket. The passenger can track the request, while station or operations staff can verify the ticket information. UI Policies and minimal Catalog Client Scripts support form behaviour, while ACLs and roles control access.
3	Novelty / Uniqueness	The solution uses ServiceNow not only as a request-management platform but as a complete low-code ticketing workflow. Its key differentiators are automated fare processing, centralized booking records, QR-based ticket generation, Service Portal-compatible QR presentation using a widget/spModal approach, and a Flow Designer-first architecture with minimal custom scripting. This makes the solution easier to maintain and hand over to ServiceNow administrators compared with a heavily custom-coded application.
4	Social Impact / Customer Satisfaction	The project can improve commuter convenience by reducing dependence on physical counters and paper tickets, providing faster self-service booking, clearer fare information, digital confirmation, and easier ticket verification. For metro operations, structured digital records and workflow automation can reduce repetitive manual work and support more consistent service. Reduced paper usage also supports environmentally responsible ticketing practices.
5	Business Model (Revenue Model)	For a real-world deployment, the system could be offered to metro or public-transport operators as an enterprise digital-ticketing solution. Possible revenue models include one-time implementation and customization fees, annual support and maintenance contracts, ServiceNow consulting/configuration charges, integration fees for payment or notification services, and future analytics/dashboard modules. In the current academic project, no commercial payment or revenue mechanism is implemented.
6	Scalability of the Solution	The solution is designed around configurable ServiceNow components such as Service Catalog, Flow Designer, custom tables, roles, ACLs, widgets, and notifications. Additional stations, routes, fare bands, passenger categories, and workflows can be added without redesigning the entire system. Future extensions can include payment gateway/UPI integration, SMS or email ticket delivery, WhatsApp-based services, travel analytics, passes, and broader multi-station deployment.